

THE NEXT MOVIE RECOMMENDATION

MovieLens 1M and adapted NextItNet

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MOVIELENS AND RATING ACTIVITY

1,000,209

ratings from 6,040 users

95.74%

of user-catalog cells unobserved

Gaps between adjacent ratings within a user

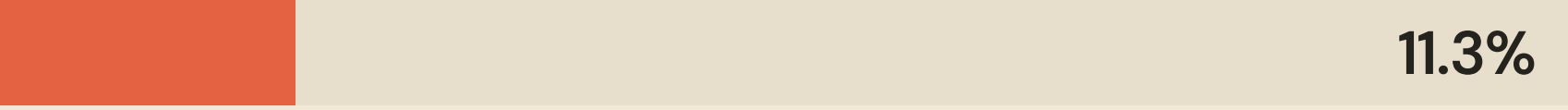
Same timestamp



1 to 60 seconds



More than 60 seconds



Rating order cannot establish viewing order.

Source: MovieLens 1M. All 994,169 within-user adjacent gaps.

COURSE METHOD AND PROJECT EXTENSION

MATRIX FACTORIZATION

Predict a missing star rating
from user and movie factors.

Lecture: softImpute with cross-validation.
Metadata can extend the prediction.

ADAPTED NEXTITNET

Predict the next rated movie
from recent movie IDs.

Causal convolution encoder.
Validation selects 10 recent items.

EVALUATION SETUP

TRAIN

988,129

Earlier ratings
Fit model candidates

VALIDATION

6,040

One per user
Choose settings

TEST

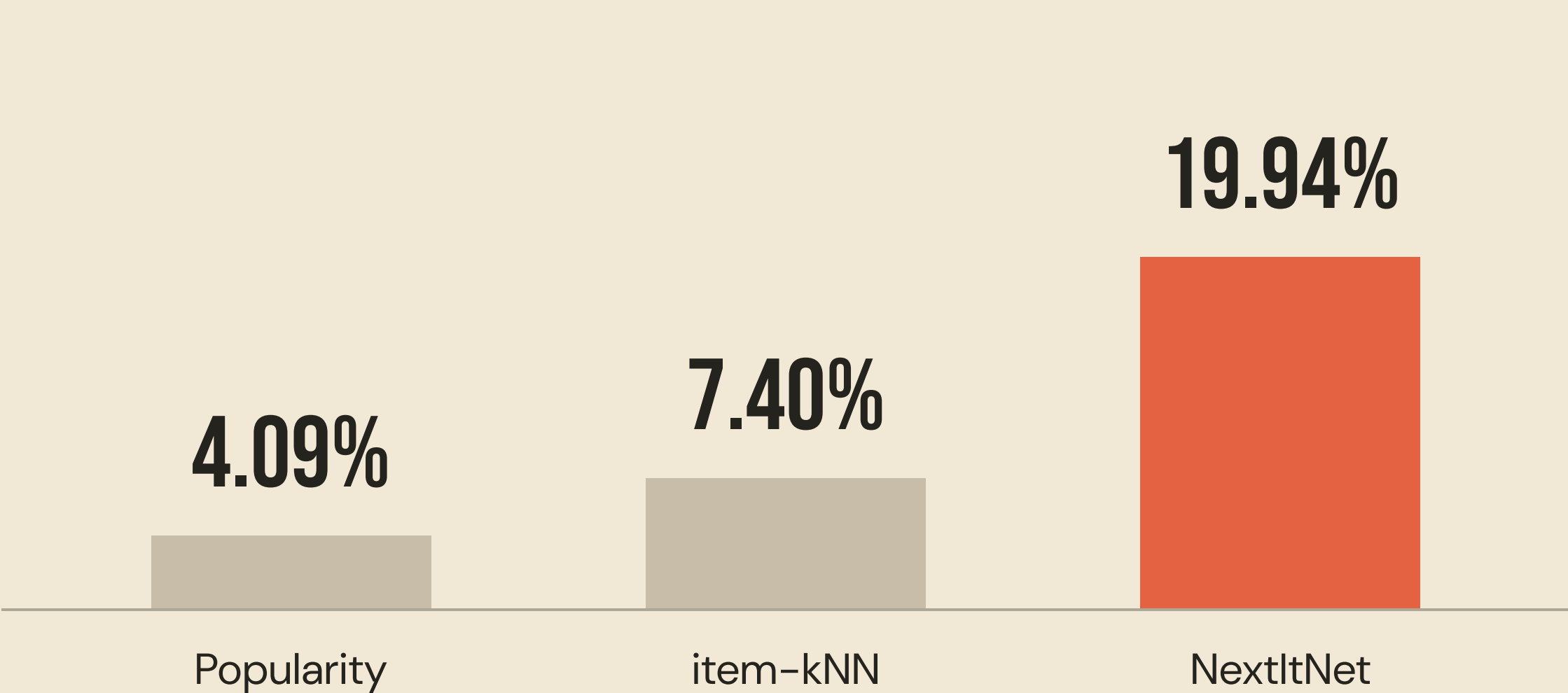
6,040

Final rating per user
Evaluate once

Same 6,038 eligible users and 3,704 fitted catalog items.
Remove previously rated movies. Two unseen targets excluded.

Final refit includes validation history. Target is the next rated movie, at any star value.

NEXTITNET LEADS THE TESTED RANKING BASELINES



+12.54

percentage points
over item-kNN

NDCG@10: 0.1049
Paired 95% improvement interval
[0.05981, 0.07224]

A STRICTER TIMESTAMP CHECK

HR@10 for 3,494 eligible users

NextItNet

14.31%

item-kNN

5.95%

Final rating timestamp is strictly later
than the preceding rating.

This cohort removes a test-boundary tie. It still contains other timing ambiguities.

The advantage persists.

A global time split and retrained
shuffled-history comparison
remain necessary.

A LIMITED PILOT IS THE NEXT STEP

MOVIE STARTS

Recommendation-led starts
per exposed user.

MovieLens cannot establish ROI. A live test needs exposure and business outcome data.
Minimize stored history, audit subgroups and provide an opt-out.

Before the pilot

Global time split and ranking-trained MF.

Serving

Popular fallback and metadata for cold start.

Guardrails

Completion, catalog concentration, latency.